

Exact Fixation and Green Kernels for a Finite Moran Process

Route-A source-local certificate HCS-C253

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Abstract

We close an exact finite-state atlas for a continuous-time Moran birth–death process with selection ratio ρ and event scale β . Fixation, occupation, absorption time, and reversible weights are rational for rational inputs. This is source-local probability; no arithmetic origin or spectral operator is claimed.

1 Model

Let $X_t \in \{0, \dots, N\}$ count one type in a fixed population. For $1 \leq i \leq N-1$, set

$$\lambda_i = \beta \rho \frac{i(N-i)}{N}, \quad \mu_i = \beta \frac{i(N-i)}{N}. \quad (1)$$

The endpoints 0 and N absorb. The transient generator Q has $Q_{i,i+1} = \lambda_i$, $Q_{i,i-1} = \mu_i$, and $Q_{ii} = -(\lambda_i + \mu_i)$.

2 Fixation theorem

Let u_i be the probability of absorption at N . The backward equation is

$$\lambda_i(u_{i+1} - u_i) + \mu_i(u_{i-1} - u_i) = 0, \quad u_0 = 0, \quad u_N = 1.$$

Theorem 1 (selection-ratio fixation). *For $\rho \neq 1$, $u_i = (1 - \rho^{-i})/(1 - \rho^{-N})$; for $\rho = 1$, $u_i = i/N$.*

Proof. Successive differences have ratio ρ^{-1} ; summing the geometric progression and taking the neutral limit gives the formula. Positivity of all transient rates and finiteness of the state space imply absorption almost surely. $\square$

3 Green kernel and time

Define $G = (-Q)^{-1}$ and $t_i = \sum_{j=1}^{N-1} G_{ij}$. Then G_{ij} is expected occupation time and

$$\lambda_i(t_{i+1} - t_i) + \mu_i(t_{i-1} - t_i) = -1, \quad t_0 = t_N = 0. \quad (2)$$

All entries are rational for rational (N, ρ, β) . The factorization separates population selection from the arbitrary event-time scale, so the Green kernel is not a fitted schedule.

4 Reversibility and boundaries

With $w_1 = 1$ and

$$\frac{w_{i+1}}{w_i} = \rho \frac{i(N-i)}{(i+1)(N-i-1)}, \quad (3)$$

one has $w_i \lambda_i = w_{i+1} \mu_{i+1}$. The $\rho = 1$ face is neutral; $\beta = 0$ freezes the chain, and $N = 1$ has no transient block.

5 Executable certificate

The producer emits eight exact rows, including every Green matrix through $N = 10$. The independent checker closes 220 assertions; SymPy closes ten identities; replay is byte-identical and hostile mutation rejects 23/23. Three fixed-epoch manuscript rounds use embedded fonts and a content-addressed manifest. Decimal values are presentation checks only. Every rational recurrence is checked before decimal conversion, including the neutral limit and the no-inversion boundary faces.

Table 1: Receipt coverage.

rows	sizes	selection	recorded object
2	3,8	neutral	fixation, Green, time
4	4,6,7,10	mixed	fixation, full Green
2	5,9	weak/strong	fixation, full Green

6 Route-A boundary

The strict evaluator tuple is (A0_FAIL,A1_PASS_ANALYTIC,A2_FAIL,A3_FAIL,A4_FORMAL_HINT). The arithmetic origin is none; no primitive target orbit or determinant convention is defined. The verdict is ROUTE_A_REJECTED, with route_b_invocation_allowed: false, under NO_BAD_EULER_OR_ROOT_NUMBER. Fixation is a population probability, not a target arithmetic matching law.

7 Conclusion

The finite Moran generator supplies both a closed selection formula and an exact occupation kernel. Explicit frozen and singleton faces preserve the A0 stopping boundary.

References

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- [2] W. J. Ewens, *Mathematical Population Genetics*, Springer, 2004.
- [3] J. R. Norris, *Markov Chains*, Cambridge Univ. Press, 1997.